Markov Decision Processes:
Reinforcement Learning - Lecture 2

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PUCRS

Material adapted from:
Sebastian Thrun - Stanford University/Udacity
Sutton and Barto - Reinforcement Learning: An Introduction (2nd ed.)
David Silver - Reinforcement Learning

Outline

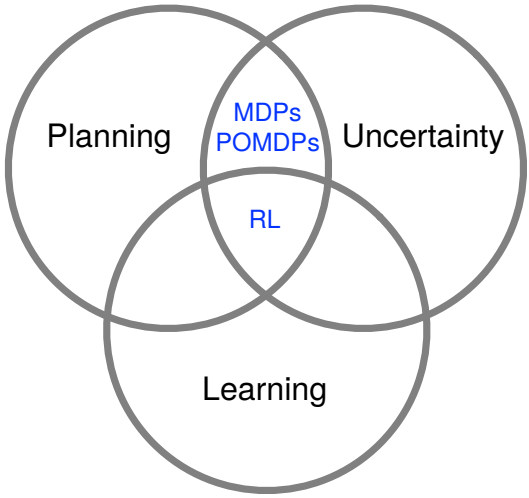
Context

- This lecture (Sutton and Barto 3.1-3.6)
  - Markov Decision Processes
  - Value functions
- Next lecture:
  - **Planning by dynamic programming**
  - Solving a **known** MDP

Recap

- We know how to get from a current state to a goal state
  - Via domain specific search
  - Via domain independent planning
- We also know how to calculate the probability of future events via probability theory
- But, how can we get the agent to decide between different actions, particularly in the face of uncertainty?

Planning Under Uncertainty



Recap

	Deterministic	Stochastic
Fully Observable		
Partially Observable		

Recap

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Partially Observable		POMDP

MDPs

- **Markov Decision Processes** (MDPs) formally describe environments for reinforcement learning
- Assumptions behind MDP environments
  - Fully observable states (states completely characterizes the process)
  - Stochastic transitions
  - Agent has **partial control** over transitions
- Almost all RL problems can be formalised as MDPs, e.g.:
  - Optimal control primarily deals with continuous MDPs
  - Partially observable problems can be converted into MDPs
  - Bandits are MDPs with one state

Outline

How Good is an action

- Consider an agent capable of planning and executing different actions to achieve different goals
- Example you can either:
  - Stay home and study
  - Go out and have fun
- Which should you do, and why?

Rationality

- A rational agent should pursue the goal that is, in some sense, "best for it"
- How can we identify whether the achievement of one goal is "better" for the agent than another?

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    Winning the lottery $U(w) = 1000$ Failing AI: $U(f) = -10$
  - But what if actions have uncertain effects?
    Buying a lottery ticket does not give us a utility of 1000

Expected Utility

- A nondeterministic action A could lead to several possible outcomes, e.g. $Result_1(A), Result_2(A), \ldots$
  $Result(BuyTicket) \in \{win, \neg win\}$
- We can associate a probability with the different outcomes

$$\mathbb{P}\left[Result_i(A) | Do(A), E\right]$$

  - E captures the evidence we have about the world
  - $Do(A)$ is the proposition that action A is executed in the current state
- The **expected utility** of executing A given E is then

$$\mathbb{E}\left[U(A) \mid E\right] = \sum_i \mathbb{P}\left[Result_i(A) \mid Do(A), E\right] U(Result_i(A))$$

- Principle of **maximum expected utility** (MEU):
  A rational agent should choose the action that maximises its expected utility

More generally

- We can use the expectation operator ($\mathbb{E}[X]$) for **any random variable** X

$$\mathbb{E}[X] \doteq \sum_{x \in X} \mathbb{P}[x]\, x$$

- Example:
  Expected value of a six sided die:

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- Example:
  Expected value of a six sided die:

x	P(X=x)	x*P(X=x)
1	1/6	1/6
2	1/6	2/6
3	1/6	3/6
4	1/6	4/6
5	1/6	5/6
6	1/6	6/6
	Total	21/6 = 3.5

Maximum Expected Utility

- An agent following the principle of MEU is rational, acting optimally in any situation
- But choosing the best sequence of actions requires enumerating all action sequences; infeasible for long sequences
- Computations are often too expensive:
  - $\mathbb{P}[Result_i(A)|Do(A), E]$ requires a complete causal model of the world. Even if all information is available, this is computationally intractable
  - Computing $U(Result_i(A))$ requires searching or planning as we do not know how good a state is until we know what goals we can reach from it (and how expensive they are)

Maximum Expected Utility II

- MEU is no silver bullet for the problems of AI
  - But does give us a nice framework for building an agent,
  - An MEU maximising agent will achieve the highest possible performance averaged over all environments in which the agent can be placed
  - This moves from a performance measure in the external environment, to an internal criteria of utility maximisation

Lotteries

- A lottery is a probability distribution over a set of actual outcomes
- A lottery L with possible outcomes $C_1, \ldots, C_n$ and associated probabilities $p_1, \ldots, p_n$ is written as

$$L = [p_1, C_1; \ldots p_n, C_n]$$

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- Example: There is a 30% likelihood I will go out tonight.
If I do so, there is a 60% likelihood I will do badly in my test.
If I do not go out, I will do well in my test with a 90% likelihood

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- Our goal is to understand the relationships between preferences between lotteries and preferences between states

- **Maximum Expected Utility Principle**: The utility of a lottery is the sum of the probability of each outcome times the utility of that outcome

$$U([p_1, S_1; \ldots, p_n, S_n]) = \sum_i p_i U(S_i)$$

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- Since outcomes of nondeterministic actions are lotteries, we obtain the MEU decision rule
  - We now know what a utility function is
  - But without structure, there is little we can do with it

Utility Functions

- A typical use of utility functions (stemming from economic theory) maps the amount of money an agent has to its utility
- Agents prefer having more money to less: such a preference is **monotonic**
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- You are a participant in a game show. You are in the final round, and can take \$1,000,000, or gamble it on a coin flip.
  If the coin lands on heads, you will win \$3,000,000, tails, you will take home nothing.
  What do you do?

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  If the coin lands on heads, you will win \$3,000,000, tails, you will take home nothing.
  What do you do?
- The **expected monetary value** (EMV) is \$1,500,000;
  this is more than the prize
- If \$1=1 utility, then a rational agent would take the gamble

- Instead, let us denote the state of having $\$n$ as S_n, and the your initial states as S_k. Then

$$EU(flip) = 0.5U(S_k) + 0.5U(S_{k+3000000}) \qquad EU(noflip) = U(S_{k+1000000})$$

- We now need to assign utilities to the different outcome states
  - Utility is typically not proportional to money:
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  - What are the expected utilities? $EU(flip) =?$ $EU(noflip) =?$

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  - Then $EU(flip) = 7.5, EU(noflip) = 8$, and you would not flip the coin
- Typically, for some lottery L, $U(L) < U(S_{EMV(L)})$. In other words, being given the cash without gambling is better than facing a lottery
  - Agents with this type of utility function are **risk averse**
  - If $U(S_{EMV(L)}) < U(L)$ the agent is **risk seeking**

Utility Functions

- The amount an agent will accept instead of a lottery is called the **certainty equivalent** of the lottery
- The difference between the expected monetary value of a lottery and its certainty equivalent is called the **insurance premium**
- The insurance premium is only positive if an agent is risk averse
- If a utility curve is linear (i.e. $U(S_{EMV(L)}) = U(L)$), the agent is

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- The insurance premium is only positive if an agent is risk averse
- If a utility curve is linear (i.e. $U(S_{EMV(L)}) = U(L)$), the agent is **risk neutral**
- Any affine transformation of a utility function is a utility function

Outline

Markov Property

The future is independent of the past, given the present

Definition

A state S_t is **Markov** if and only if

$$\mathbb{P}\left[S_{t+1} \mid S_t\right] = \mathbb{P}\left[S_{t+1} \mid S_1, \ldots S_t\right]$$

- The state captures all relevant information from the history
- Once the state is known, the history may be thrown away
  i.e. The state is a sufficient statistic of the future

State Transition Matrix

For a Markov state s and successor state s', the **state transition probability** is defined by

$$\mathcal{P}_{ss'} = \mathbb{P}\left[S_{t+1} = s' \mid S_t = s\right]$$

State transition matrix $\mathcal{P}$ defines transition probabilities from all states s to all successor states s',

$$\mathcal{P} = \begin{array}{c} \\ \textit{from} \end{array} \begin{array}{c} \begin{array}{ccc} & \textit{to} & \end{array} \\ \begin{bmatrix} \mathcal{P}_{11} & \dots & \mathcal{P}_{1n} \\ \vdots & & \\ \mathcal{P}_{n1} & \dots & \mathcal{P}_{nn} \end{bmatrix} \end{array}$$

where each row of the matrix sums to 1.

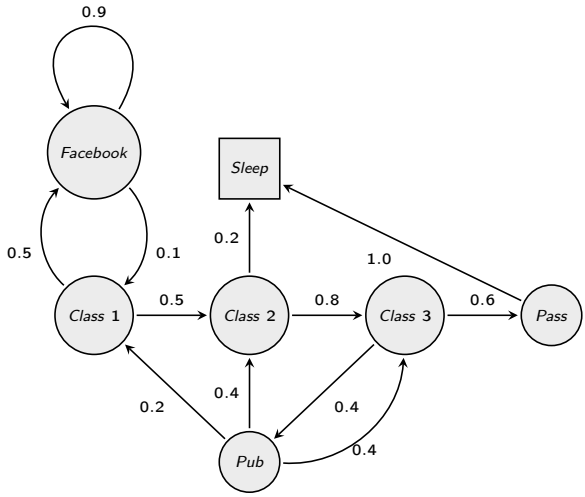
A Markov process is a memoryless random process, i.e. a sequence of random states S_1, S_2, ... with the Markov property.

Definition

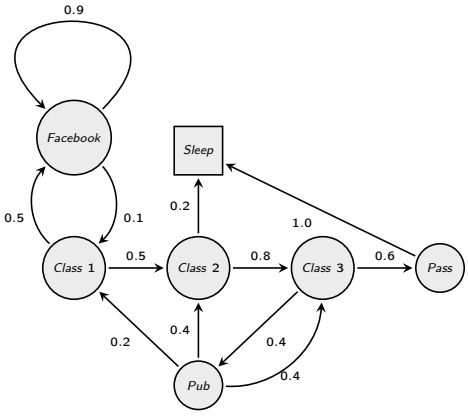
A Markov Process (or **Markov Chain**) is a tuple $\langle \mathcal{S}, \mathcal{P} \rangle$

- $\mathcal{S}$ is a (finite) set of states
- $\mathcal{P}$ is a state transition probability matrix,
  $\mathcal{P}_{ss'} = \mathbb{P}[S_{t+1} = s' \mid S_t = s]$

Example: Student Markov Chain



Example: Student Markov Chain Episodes

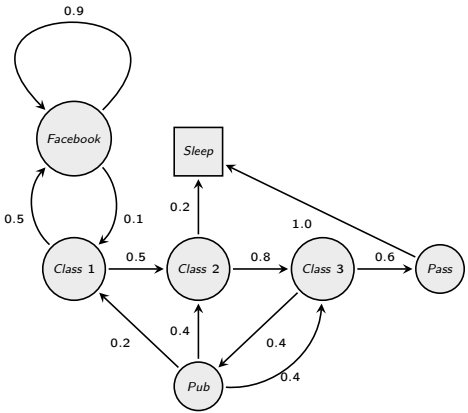


Sample **episodes** for Student Markov Chain starting from $S_1 = C1$

$$S_1, S_2, \ldots, S_T$$

- C1 C2 C3 Pass Sleep
- C1 FB FB C1 C2 Sleep
- C1 C2 C3 Pub C2 C3 Pass Sleep
- C1 FB FB C1 C2 C3 Pub C1 FB FB FB C1 C2 C3 Pub C2 Sleep

Example: Student Markov Chain Transition Matrix



$$\mathcal{P} = \begin{array}{c} \\ C1 \\ C2 \\ C3 \\ Pass \\ Pub \\ FB \\ Sleep \end{array} \begin{array}{c} \begin{array}{ccccccc} C1 & C2 & C3 & Pass & Pub & FB & Sleep \end{array} \\ \begin{bmatrix} & 0.5 & & & & 0.5 & \\ & & 0.8 & & & & 0.2 \\ & & & 0.6 & 0.4 & & \\ & & & & & & 1.0 \\ 0.2 & 0.4 & 0.4 & & & & \\ 0.1 & & & & & 0.9 & \\ & & & & & & 1.0 \end{bmatrix} \end{array}$$

Outline

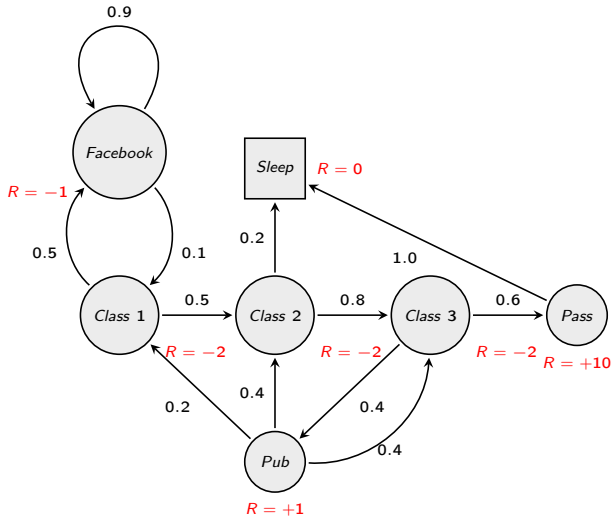
A Markov Reward Processes is a Markov chain with values

Definition

A **Markov Reward Process** is a tuple $\langle \mathcal{S}, \mathcal{P}, \mathcal{R}, \gamma \rangle$

- $\mathcal{S}$ is a finite set of states
- $\mathcal{P}$ is a transition probability matrix/function, $\mathcal{P}_{ss'} = \mathbb{P}[S_{t+1} = s' \mid S_t = s]$
- $\mathcal{R}$ is a reward function, $R_s = \mathbb{E}[R_{t+1} \mid S_t = s]$
- γ is a discount factor, $\gamma \in [0, 1]$

Example: Student MRP



Definition

The return G_t is the total discounted reward from time-step t.

$$G_t = R_{t+1} + \gamma R_{t+2} + \cdots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

- The discount $\gamma \in [0, 1]$ is the present value of the future rewards
- The value of receiving reward R after $k + 1$ time steps is $\gamma^k R$
- This values immediate reward above delayed reward
  - γ close to 0 leads to "myopic" evaluation
  - γ close to 1 leads to "far-sighted" evaluation

Why discount?

Most Markov reward and decision processes are discounted. Why?

- Mathematically convenient to discount rewards
- Avoids infinite returns in cyclic Markov processes
- Uncertainty about the future may not be fully represented
- If the reward is financial, immediate rewards may earn more interest than delayed rewards
- Animal/human behaviour shows preference for immediate reward
- It is sometimes possible to use undiscounted Markov reward processes (i.e. $\gamma = 1$), e.g. if all sequences terminate.

Value function

The value function $v(s)$ gives the long-term value of state s

Definition

The **state value function** $v(s)$ of an MRP is the expected return starting from state s

$$v(s) = \mathbb{E}\left[G_t \mid S_t = s\right]$$

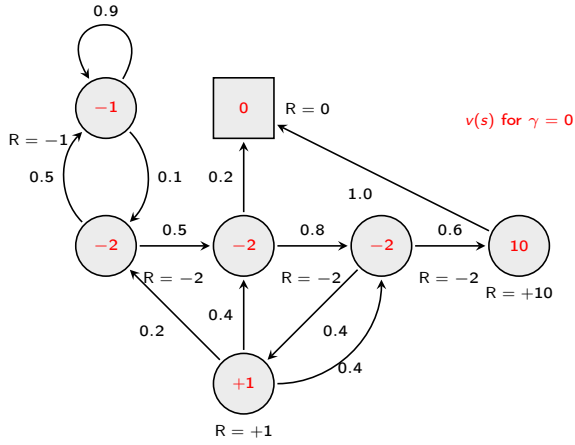
Example: Student MRP Returns

Sample **returns** for Student MRP: Starting from $S_1 = C1$ with $\gamma = \frac{1}{2}$

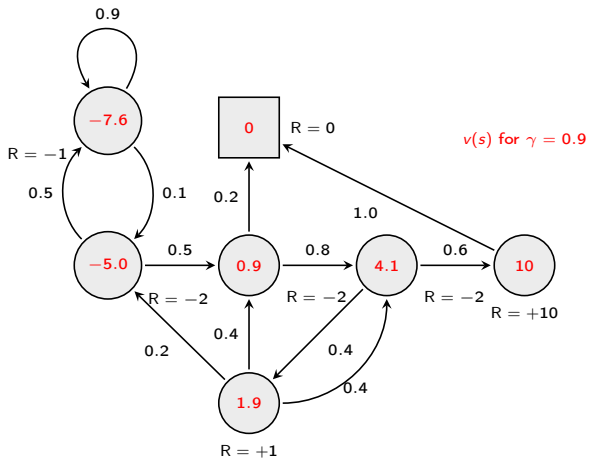
$$G_1 = R_2 + \gamma R_3 + \cdots + \gamma^{T-2} R_T$$

C1 C2 C3 Pass Sleep	$v_1 = -2 - 2 * \frac{1}{2} - 2 * \frac{1}{4} + 10 * \frac{1}{8} = -2.25$
C1 FB FB C1 C2 Sleep	$v_1 = -2 - 1 * \frac{1}{2} - 1 * \frac{1}{4} + 2 * \frac{1}{8} - 2 * \frac{1}{16} = -3.125$
C1 C2 C3 Pub C2 C3 Pass Sleep	$v_1 = -2 - 2 * \frac{1}{2} - 2 * \frac{1}{4} + 1 * \frac{1}{8} - 2 * \frac{1}{16} = -3.41$
C1 FB FB C1 C2 C3 Pub C1 ...	$v_1 = -2 - 1 * \frac{1}{2} - 1 * \frac{1}{4} - 2 * \frac{1}{8} - 2 * \frac{1}{16} = -3.20$
FB FB FB C1 C2 C3 Pub C2 Sleep	

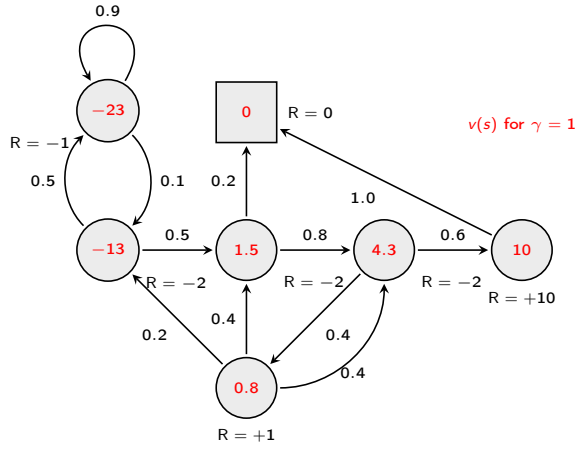
Example: State-Value Function for Student (MRP) (1)



Example: State-Value Function for Student (MRP) (2)



Example: State-Value Function for Student (MRP) (3)



Bellman Equation for MRPs

The value function can be decomposed into two parts:

- immediate reward R_{t+1}
- discounted value of successor state $\gamma\, v(S_{t+1})$

$$v(s) = \mathbb{E}\left[G_t \mid S_t = s\right]$$

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&= \mathbb{E}\left[R_{t+1} + \gamma (R_{t+2} + \gamma R_{t+3} + \ldots) \mid S_t = s\right] \\
&= \mathbb{E}\left[R_{t+1} + \gamma G_{t+1} \mid S_t = s\right] \\
&= \mathbb{E}\left[R_{t+1} + \gamma v(S_{t+1}) \mid S_t = s\right]
\end{aligned}
$$

Bellman Equation for MRPs (2)

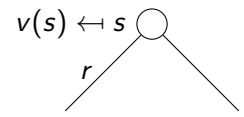
$$v(s) = \mathbb{E}\left[? \mid S_t = s\right]$$

$$v(s) \leftarrow\!\shortmid\ s \bigcirc$$

$$v(s) = ?$$

Bellman Equation for MRPs (2)

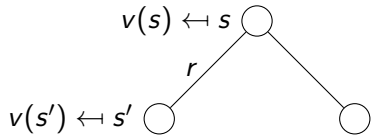
$$v(s) = \mathbb{E}\left[\mathrm{R}_{t+1} \mid \mathrm{S}_t = s\right]$$



$$v(s) = \mathcal{R}_s$$

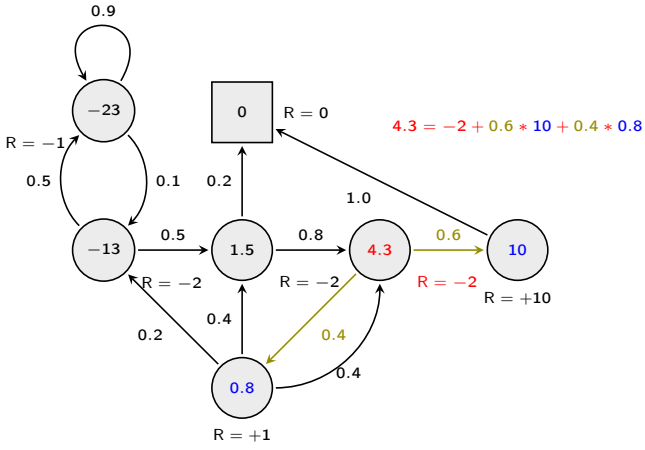
Bellman Equation for MRPs (2)

$$v(s) = \mathbb{E}\left[R_{t+1} + \gamma\, v(S_{t+1}) \mid S_t = s\right]$$



$$v(s) = \mathcal{R}_s + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}\, v(s')$$

Example: Bellman Equation for Student MRP



Bellman Equation in Matrix Form

The Bellman equation can be expressed concisely using matrices,

$$v = \mathcal{R} + \gamma \mathcal{P}\, v$$

where v is a column vector with one entry per state

$$\begin{bmatrix} v(1) \\ \vdots \\ v(n) \end{bmatrix} = \begin{bmatrix} \mathcal{R}_1 \\ \vdots \\ \mathcal{R}_n \end{bmatrix} + \gamma \begin{bmatrix} \mathcal{P}_{11} & \dots & \mathcal{P}_{1n} \\ \vdots & & \\ \mathcal{P}_{n1} & \dots & \mathcal{P}_{nn} \end{bmatrix} \begin{bmatrix} v(1) \\ \vdots \\ v(n) \end{bmatrix}$$

Solving the Bellman Equation

- The Bellman equation is a linear equation
- It can be solved directly:

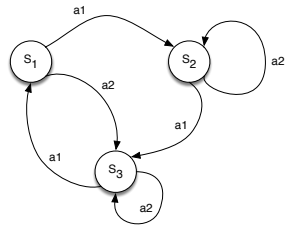
$$
\begin{aligned}
v &= \mathcal{R} + \gamma \mathcal{P} v \\
(\mathrm{I} - \gamma \mathcal{P}) v &= \mathcal{R} \\
v &= (\mathrm{I} - \gamma \mathcal{P})^{-1} \mathcal{R}
\end{aligned}
$$

- Computational complexity is $O(n^3)$ for n states
- Direct solution only possible for small MRPs
- There are many iterative methods for large MRPs, e.g.
  - Dynamic programming
  - Monte-Carlo evaluation
  - Temporal-Difference learning

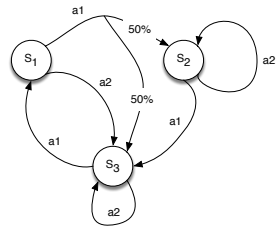
Outline

Markov Decision Process (MDP)

- Such a model, in a fully observable environment, is called a **Markov Decision Process**



Markov Decision Process (MDP)



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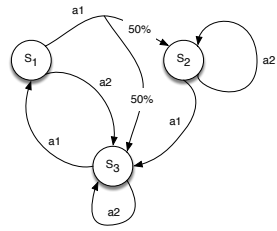
- An MDP is defined in terms of
  1. An initial state S_0
  2. A transition model $T(s, a, s')$
  3. A reward function $R(s)$

Markov Decision Process (MDP)



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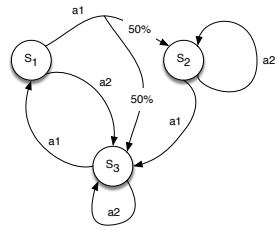
Markov Decision Process (MDP)



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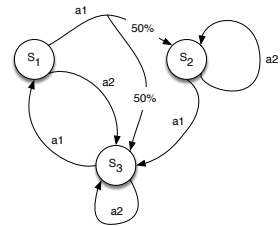
Markov Decision Process (MDP) - In Sutton and Barto's Book



- Such a model, in a fully observable environment, is called a **Markov Decision Process**
- An MDP is defined in terms of:
  1. A finite set of states $\mathcal{S}$
  2. A finite set of actions $\mathcal{A}$
  3. A markovian transition model $T(s, a, s') = \mathbb{P}(S_{t+1} = s' \mid S_t = s, A_t = a)$
  4. A reward function $R(s) = \mathbb{E}[R_{t+1} \mid S_t = s, A_t = a]$
  5. A discount factor $\gamma \in [0, 1]$
- A solution to a MDP must specify what the agent should do for any state. Such a solution is called a **policy**

A Markov decision process (MDP) is a Markov reward process with decisions. It is an environment in which all states are Markov.

Definition

A Markov Decision Process is a tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$:

1. A finite set of states $\mathcal{S}$
2. A finite set of actions $\mathcal{A}$
3. A markovian transition model $\mathcal{P}^{a}_{ss'} = \mathbb{P}(S_{t+1} = s' \mid S_t = s, A_t = a)$
4. A reward function $\mathcal{R}^{a}_{s} = \mathbb{E}[R_{t+1} \mid S_t = s, A_t = a]$
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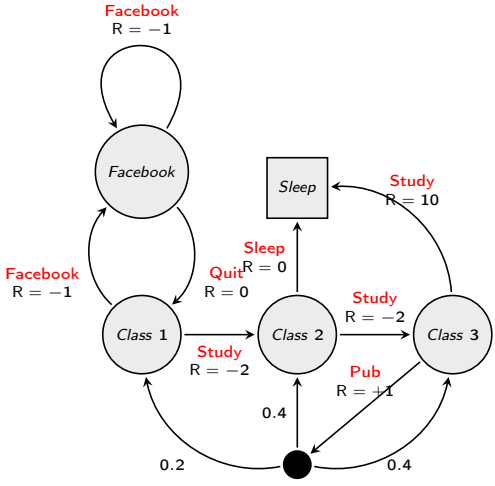
Student MDP Example



Deterministic Policies

- We denote a policy by π
- $\pi(s)$ identifies the action recommended by the policy in state s
- A complete policy tells an agent what to do no matter the outcome of an action
  - It explicitly represents the agent function and therefore describes a simple reflex agent, computed from the information used for a utility based agent

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  - The quality of a policy is measured by the **expected** utility of the possible environment histories generated by executing that policy
  - An **optimal policy** is one that yields the highest expected utility, and is denoted π^*

Stochastic Policies (1)

Definition

A **stochastic** policy π is a distribution over actions given states,

$$\pi(a \mid s) = \mathbb{P}\left[A_t = a \mid S_t = s\right]$$

- A policy fully defines the behaviour of an agent
- MDP policies generally depend on the current state (not the history)
- i.e. Policies are stationary (time-independent),
  $A_t \sim \pi(\cdot \mid S_t), \forall t > 0$

Stochastic Policies (2)

- Given an MDP $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ and a policy π
- The state sequence $S_1, S_2, \ldots$ is a Markov Process $\langle \mathcal{S}, \mathcal{P}^\pi \rangle$
- The state and reward sequence $S_1, R_2, S_2, \ldots$ is a Markov Reward Process $\langle \mathcal{S}, \mathcal{P}^\pi, \mathcal{R}^\pi, \gamma \rangle$, where

$$\mathcal{P}^\pi_{ss'} = \sum_{a \in \mathcal{S}} \pi(a \mid s)\, \mathcal{P}^a_{ss'}$$

$$\mathcal{R}^\pi_s = \sum_{a \in \mathcal{S}} \pi(a \mid s)\, \mathcal{R}^a_s$$

Value Functions (Sutton and Barto)

Definition

The **state-value function** $v_\pi(s)$ of an MDP is the expected return starting from state s, and then following policy π

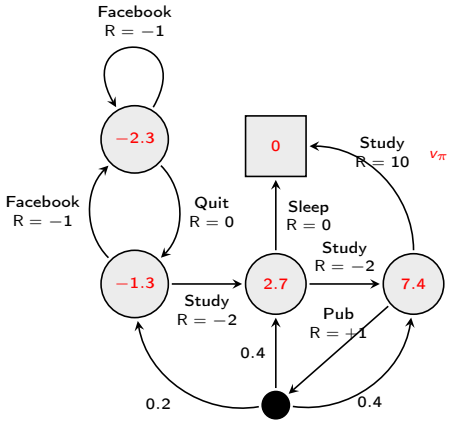
$$v_\pi(s) = \mathbb{E}_\pi \left[G_t \mid S_t = s \right]$$

Definition

The **action-value function** $q_\pi(s, a)$ is the expected return starting from state s, taking action a and then following policy π

$$q_\pi(s, a) = \mathbb{E}_\pi \left[G_t \mid S_t = s, A_t = a \right]$$

Example: State-Value Function for Student MDP



Bellman Expectation Equation

The state-value function can again be decomposed into immediate reward plus discounted value of successor state,

$$v_\pi(s) = \mathbb{E}_\pi \left[G_t \mid S_t = s \right]$$

The action-value function can similarly be decomposed,

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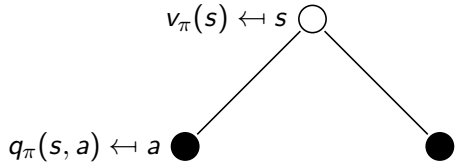
$$v_\pi(s) = \mathbb{E}_\pi \left[R_{t+1} + \gamma \, v_\pi(S_{t+1}) \mid S_t = s \right]$$

The action-value function can similarly be decomposed,

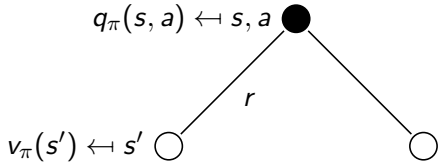
$$q_\pi(s, a) = \mathbb{E}_\pi \left[R_{t+1} + \gamma \, q_\pi(S_{t+1}, A_{t+1}) \mid S_t = s, A_t = a \right]$$

Bellman Expectation Equation for v_π



$$v_\pi(s) = \sum_{a \in \mathcal{A}} \pi(a \mid s) q_\pi(s, a)$$

Bellman Expectation Equation for Q_π



$$q_\pi(s,a) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a \, v_\pi(s')$$

Bellman Expectation Equation for v_π (2)

For deterministic policies $\pi(s)$ and state-based rewards $\mathcal{R}_s$

$$v_\pi(s) \leftarrow$$

Bellman Expectation Equation for v_π (2)

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$$v_\pi(s) \leftarrow\!\shortmid\ s \bigcirc$$

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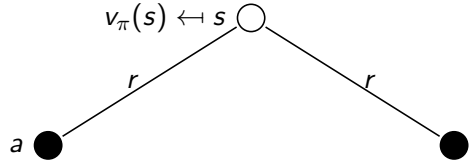
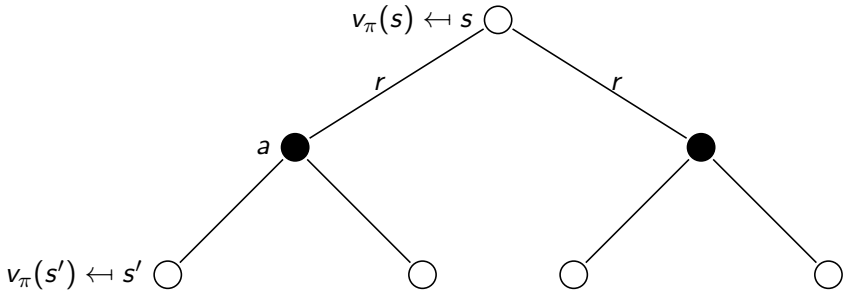


$$v_\pi(s) \leftarrow \mathcal{R}_s + [\mathcal{P}^a_{ss'}]$$

Bellman Expectation Equation for v_π (2)

For deterministic policies $\pi(s)$ and state-based rewards $\mathcal{R}_s$



$$v_\pi(s) \leftarrow \mathcal{R}_s + \left[\sum_{s'} \mathcal{P}^a_{ss'} \, v_\pi(s') \right]$$

Bellman Expectation Equation for v_π (2)

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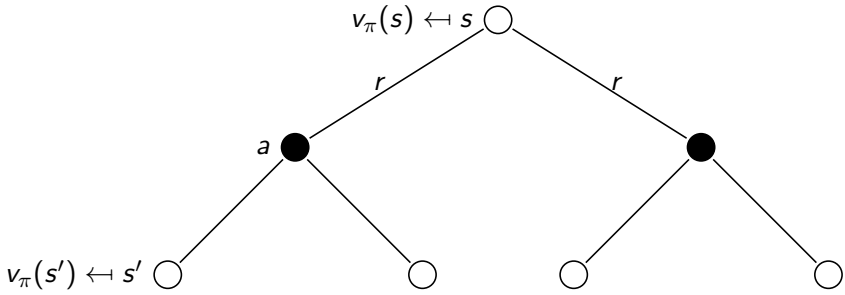

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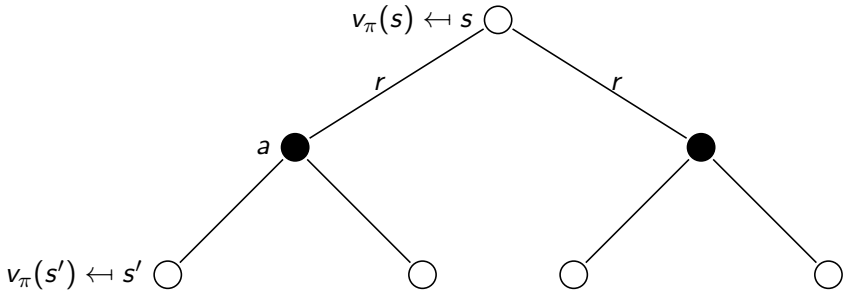
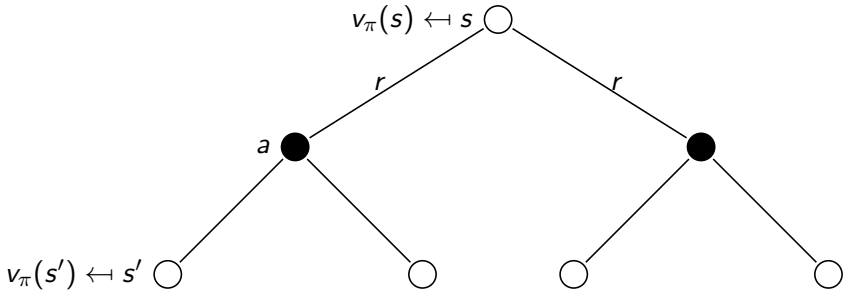


$$v_\pi(s) \leftarrow \mathcal{R}_s + \left[\gamma \sum_{s'} \mathcal{P}_{ss'}^{\pi(s)} \, v_\pi(s') \right]$$

Bellman Expectation Equation for v_π (2)

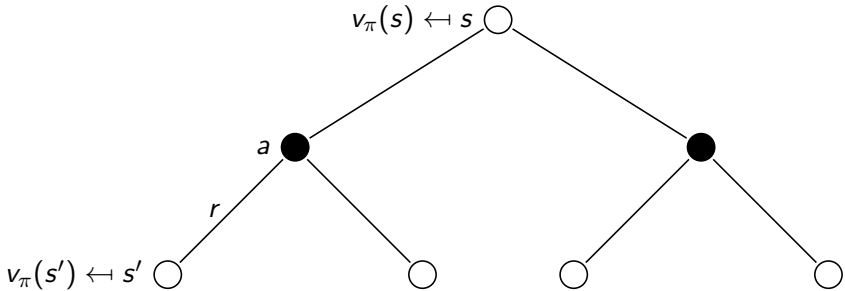
For deterministic policies $\pi(s)$ and state-based rewards $\mathcal{R}_s$



$$v_\pi(s) \leftarrow \begin{cases} \mathcal{R}_s \;\; \textit{if terminal} \\ \mathcal{R}_s + \left[\max_a \gamma \sum_{s'} \mathcal{P}_{ss'}^{\pi(s)} \, v_\pi(s')\right] \end{cases}$$

Back up function

Bellman Expectation Equation for v_π (3)



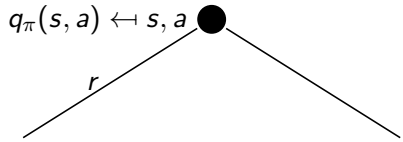
$$v_\pi(s) = \sum_{a \in \mathcal{A}} \pi(a \mid s) \left(\mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a \, v_\pi(s') \right)$$

Bellman Expectation Equation for q_π (2)

$q_\pi(s, a) \leftarrow\!\shortmid\, s, a$ 

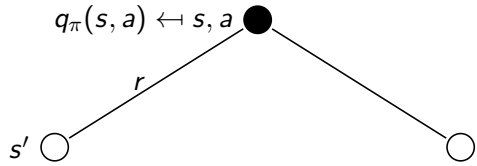
$q_\pi(s, a) = ?$

Bellman Expectation Equation for q_π (2)



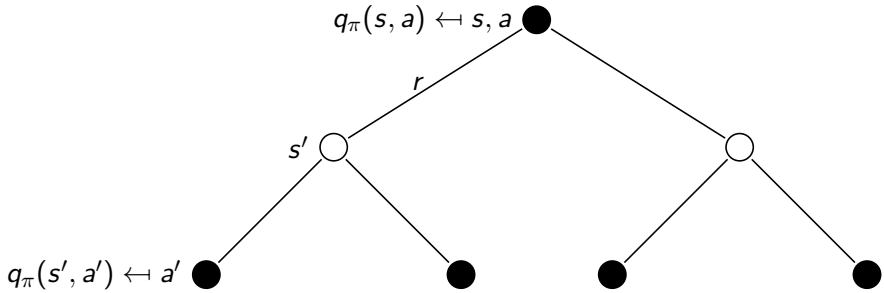
$$q_\pi(s, a) = \mathcal{R}_s^a$$

Bellman Expectation Equation for q_π (2)



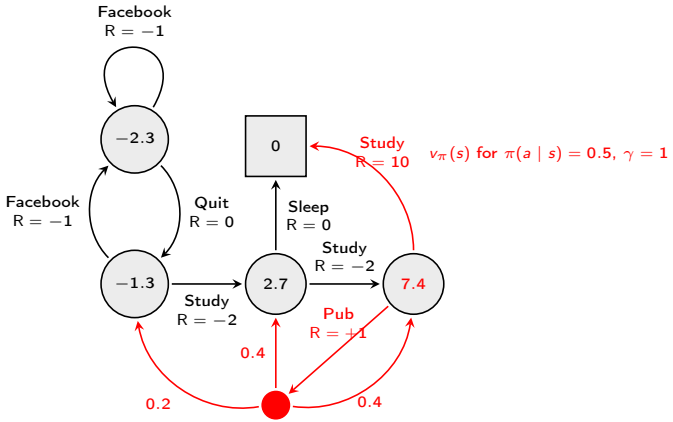
$$q_\pi(s,a) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a$$

Bellman Expectation Equation for q_π (2)



$$q_\pi(s, a) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a \sum_{a' \in \mathcal{A}} \pi(a' \mid s') q_\pi(s', a')$$

Example: Bellman Expectation Equation in Student MDP



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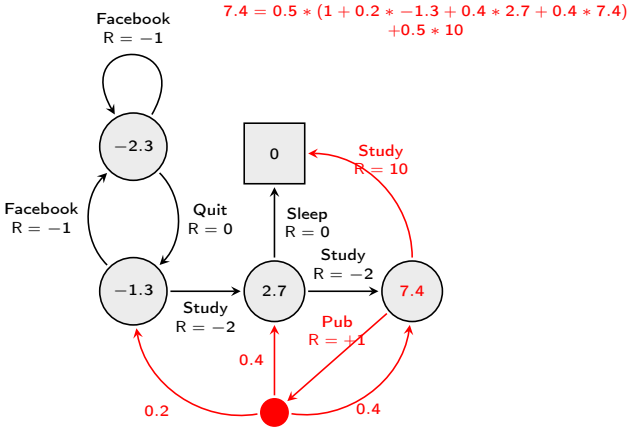


Bellman Expectation Equation (Matrix Form)

The Bellman expectation equation can be expressed concisely using the induced MRP,

$$v_\pi = \mathcal{R}^\pi + \gamma \mathcal{P}^\pi v_\pi$$

with direct solution

$$v_\pi = (\mathrm{I} - \gamma \mathcal{P}^\pi)^{-1} \mathcal{R}^\pi$$

Optimal Value Function

Definition

The optimal state-value function $v_*(s)$ is the maximum value function over all policies

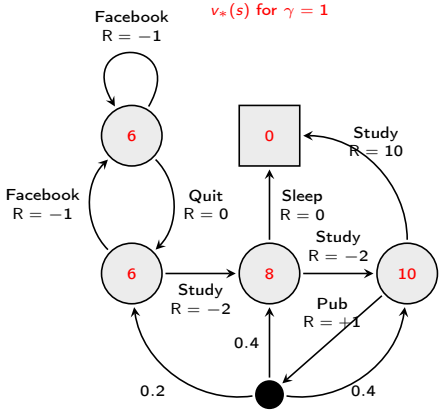
$$v_*(s) = \max_\pi v_\pi(s)$$

The optimal action-value function $q_*(s, a)$ is the maximum action-value function over all policies

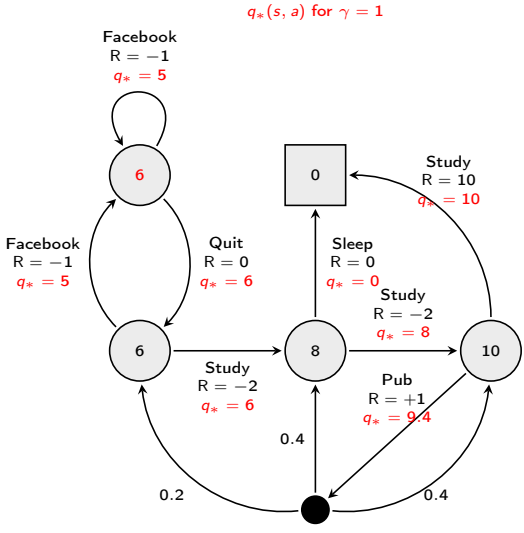
$$q_*(s, a) = \max_\pi q_\pi(s, a)$$

- The optimal value function specifies the best possible performance in the MDP.
- An MDP is "solved" when we know the optimal value fn.

Example: Optimal Value Function for Student MDP



Example: Optimal Action-Value Function for Student MDP



Define a partial ordering over policies

$$\pi \geq \pi' \text{ if } v_\pi(s) \geq v_{\pi'}(s), \forall s \in \mathcal{S}$$

Theorem

For any Markov Decision Process

- *There exists an optimal policy* π_* *that is better than or equal to all other policies,* $\pi_* \geq \pi, \forall \pi$
- *All optimal policies achieve the optimal value function,* $v_{\pi_*}(s) = v_*(s)$
- *All optimal policies achieve the optimal action-value function,* $q_{\pi_*}(s, a) = q_*(s, a)$

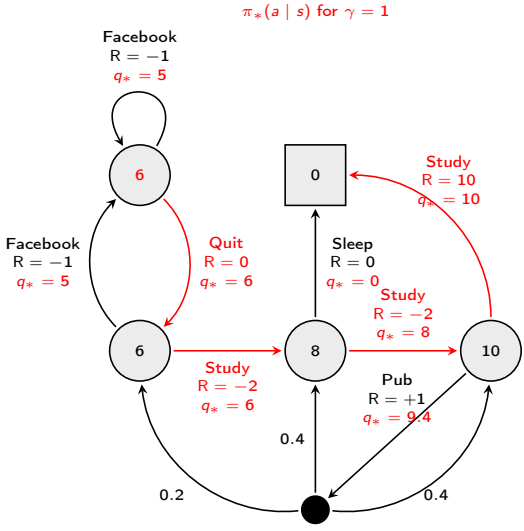
Finding an Optimal Policy

An optimal policy can be found by maximising over $q_*(s, a)$,

$$\pi_*(a \mid s) = \begin{cases} 1 & \text{if } a = \arg\max_{a \in \mathcal{A}} q_*(s, a) \\ 0 & \text{otherwise} \end{cases}$$

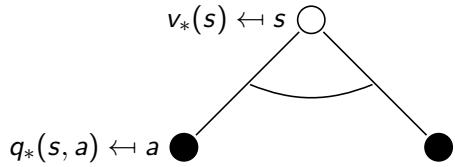
- There is always a deterministic optimal policy for any MDP
- If we know $q_*(s, a)$, we immediately have the optimal policy

Example: Optimal Policy for Student MDP



Bellman Optimality Equation for v_*

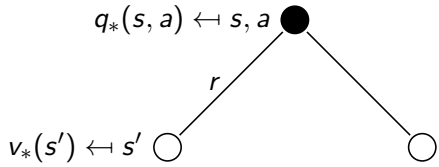
The optimal value functions are recursively related by the Bellman optimality equations:



$$v_*(s) = \max_a q_*(s,a)$$

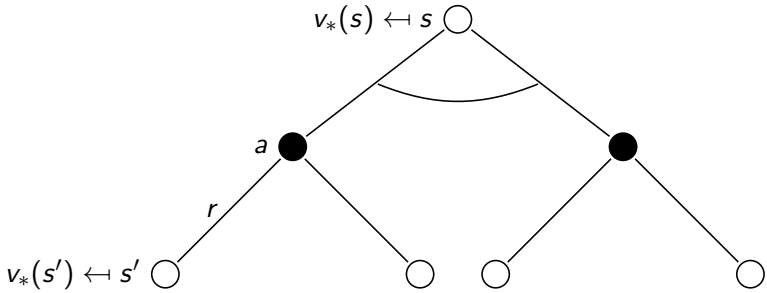
Bellman Optimality Equation for Q^*

The optimal value functions are recursively related by the Bellman optimality equations:



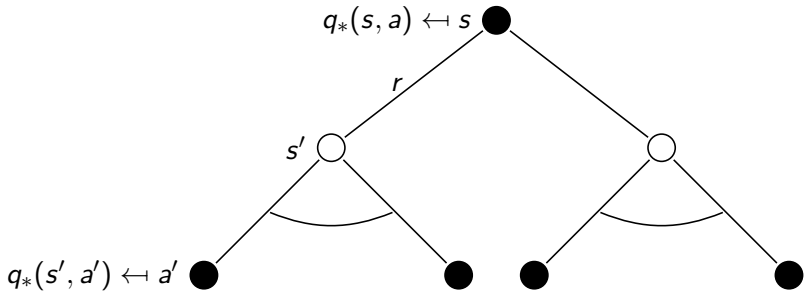
$$q_*(s, a) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a \, v_*(s')$$

Bellman Optimality Equation for V^*



$$v_*(s) = \max_a \left[\mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a \, v_*(s') \right]$$

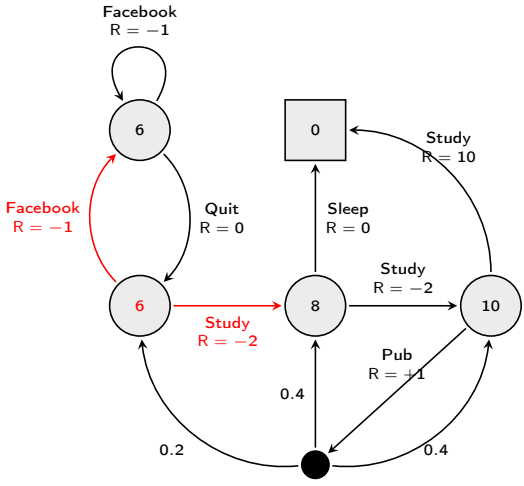
Bellman Optimality Equation for Q^*



$$q_*(s, a) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a \max_{a'} q_*(s', a')$$

Example: Bellman Optimality Equation in Student MDP

$6 = \max(-2 + 8, -1 + 6)$



Solving the Bellman Optimality Equation

- Bellman Optimality Equation is non-linear
- No closed form solution (in general)
- Many iterative solution methods
  - Value Iteration
  - Policy Iteration
- Machine Learning methods for **approximating** the Bellman Equation
  - Q-learning
  - Sarsa

Outline

Extensions to MDPs

- Infinite and continuous MDPs
- Partially observable MDPs
- Undiscounted, average reward MDPs

Infinite MDPs

The following extensions are all possible:

- Countably infinite state and/or action spaces
  - Straightforward
- Continuous state and/or action spaces
  - Closed form for linear quadratic model (LQR)
- Continuous time
  - Requires partial differential equations
  - Hamilton-Jacobi-Bellman (HJB) equation
  - Limiting case of Bellman equation as time-step $\rightarrow 0$

A Partially Observable Markov Decision Process is an MDP with hidden states. It is a hidden Markov model with actions.

Definition

A POMDP is a tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{O}, \mathcal{P}, \mathcal{R}, \mathcal{Z}, \gamma \rangle$

- $\mathcal{S}$ is a finite set of states
- $\mathcal{A}$ is a finite set of actions
- $\mathcal{O}$ is a finite set of observations
- $\mathcal{P}$ is a state transition probability matrix, $\mathcal{P}^a_{ss'} = \mathbb{P}\left[S_{t+1} = s' \mid S_t = s, A_t = a\right]$
- $\mathcal{R}$ is a reward function $\mathcal{R}^a_s = \mathbb{E}[R_{t+1} \mid S_t = s, A_t = a]$
- $\mathcal{Z}$ is an observation function, $\mathcal{Z}^a_{s'o} = \mathbb{P}\left[O_{t+1} = o \mid S_{t+1} = s', A_t = a\right]$
- A discount factor $\gamma \in [0, 1]$

POMDPs

- Current state is only **partially observable** — Partially Observable MDPs
- Planning, thus, is not done on the state of spaces
- Rather, planning is done over **belief states** — Probability distributions
- Policy then, is no longer a simple vector with a node for each state, but a multidimensional matrix with as many dimensions as there are states

Definition

A **history** H_t is a sequence of actions, observations and rewards,

$$\mathrm{H}_t = A_0, O_1, R_1, \ldots, A_{t-1}, O_t, R_t$$

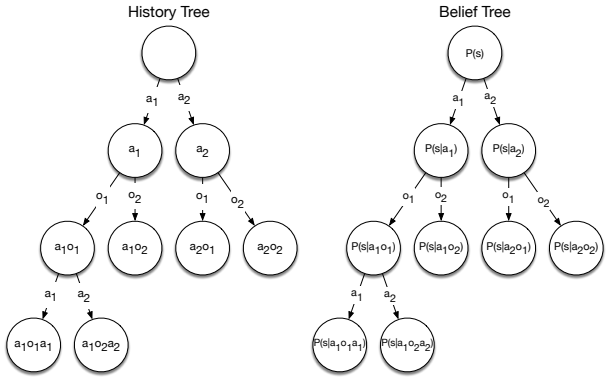
Definition

A belief state $b(h)$ is a probability distribution over states conditioned on the history h

$$b(h) = \left(\mathbb{P}\left[S_t = s^1 \mid H_t = h\right], \ldots \mathbb{P}\left[S_t = s^n \mid H_t = h\right]\right)$$

Reductions of POMDPs

- The history H_t satisfies the Markov Property
- The belief state $b(H_t)$ satisfies the Markov Property



- A POMDP can be reduced to an (infinite) history tree
- A POMDP can be reduced to an (infinite) belief state tree

Ergodic Markov Process

An ergodic Markov process is

- **Recurrent**: each state is visited an infinite number of times
- **Aperiodic**: each state is visited without any systematic period

Theorem

An ergodic Markov process has a limiting stationary distribution $\mu_\pi(\mathsf{s})$ *with the property*

$$\mu_\pi(\mathsf{s}) = \sum_{\mathsf{s}' \in \mathcal{S}} \mu_\pi(\mathsf{s}')\, \mathcal{P}_{\mathsf{s}'\,\mathsf{s}}$$

Definition

An MDP is ergodic if the Markov chain induced by any policy is ergodic.

For any policy π, an ergodic MDP has an average reward per time-step $\bar{\mathrm{R}}^{\pi}$ that is independent of start state.

$$\bar{\mathrm{R}}^{\pi} = \lim_{T \to \infty} \frac{1}{T} \mathbb{E}\left[\sum_{t=1}^{T} \mathrm{R}_t\right]$$

Average Reward Value Function

- The value function of an undiscounted, ergodic MDP can be expressed in terms of average reward.

Lecture Summary

- Stochastic Planning
- Utilities and the Expectation Operator
- Markov Processes
- Markov Reward Processes
- Markov Decision Processes
  - Formalism
  - Policies
- Extensions to MDP